

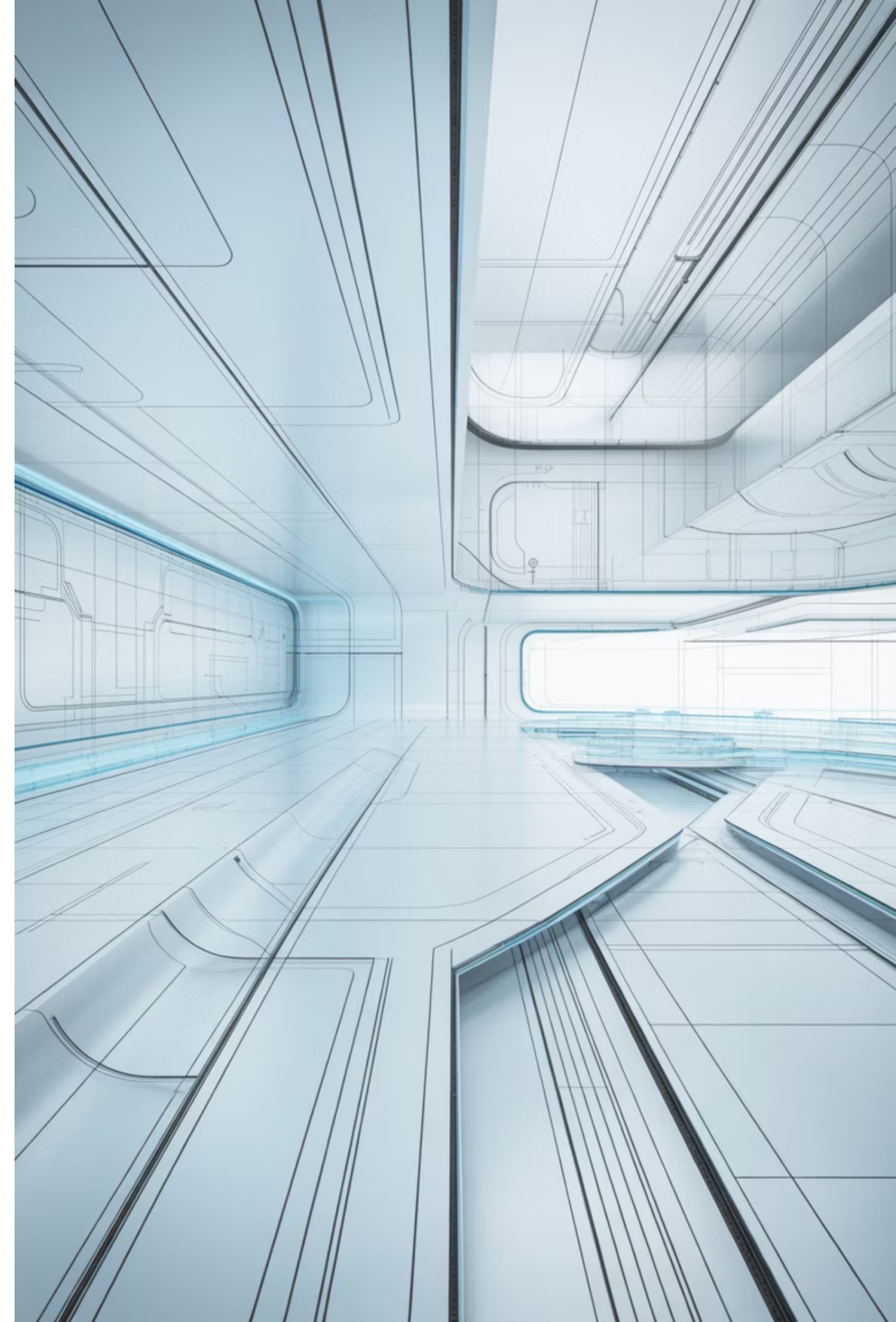


Ishaya, Jeremiah Ayock

Research Design & Methodology Development

Frameworks & Choices

This week marks a critical transition in your research journey, from reviewing existing literature to constructing the architectural foundation of your own study. Research design is the strategic blueprint that connects your research questions to your data collection methods, analytical approaches, and ultimately, your findings. It's the framework that ensures your work is valid, reproducible, and logically coherent from start to finish.



Learning Objectives



Understand Research Design Structure

Grasp the fundamental architecture of a research design, including its key components and how they interconnect to form a cohesive investigative framework.



Identify & Justify Methodological Choices

Learn to select appropriate methodologies aligned with your research questions and defend these choices with solid theoretical and practical reasoning.



Map Data Science Workflows

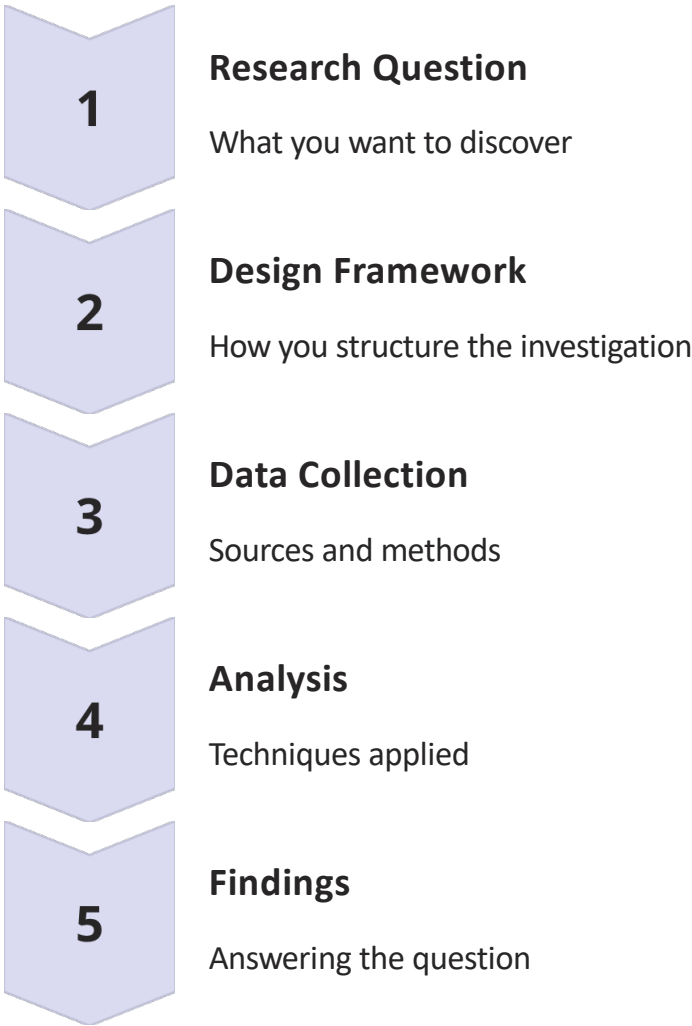
Apply established frameworks like CRISP-DM and KDD to structure thesis-level research in data science and applied fields.

By the end of this session, you will be equipped to design a research study that bridges theoretical rigor with practical execution, ensuring that every methodological decision serves your overarching research objectives.

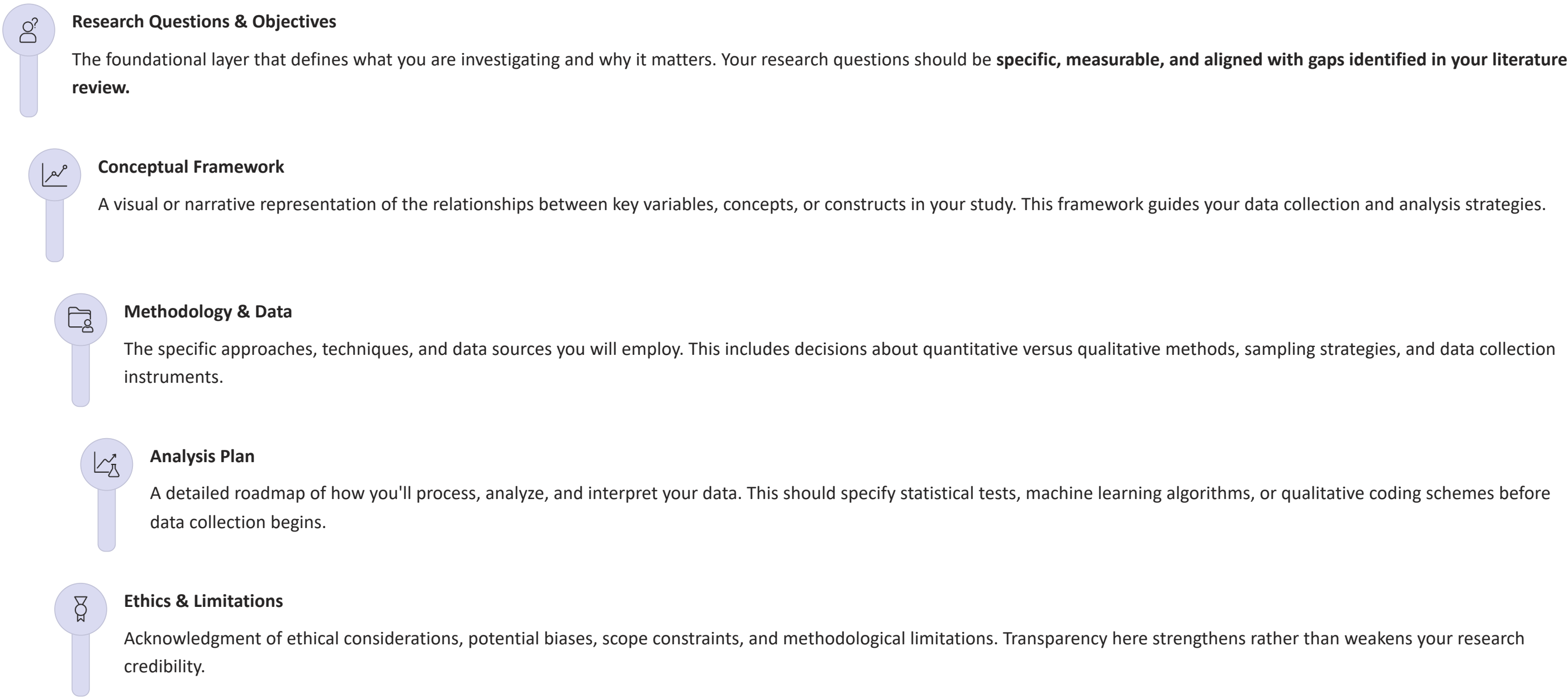
The Purpose of Research Design

Think of **research design** as the comprehensive blueprint for your entire study; it is the strategic plan that guides every decision you make throughout your research journey. A well-crafted research design serves three critical functions that distinguish rigorous academic work from mere data exploration.

First, it ensures **validity**, that your study actually measures or investigates what it claims to measure. Second, it guarantees **reproducibility**, allowing other researchers to replicate your work and verify your findings. Third, it creates **logical coherence**, ensuring that every component of your study connects meaningfully to address your research questions. Without a solid design, even the most sophisticated analytical techniques risk producing unreliable or irrelevant results.



Anatomy of a Good Research Design



Each component builds upon the previous one, creating a structured hierarchy that transforms a broad research interest into a concrete, executable study. The strength of your design lies not in complexity but in the clarity and alignment of these five essential elements.

Quantitative, Qualitative, and Mixed Methods

Understanding the fundamental methodological approaches available to researchers is essential for making informed design decisions. Each approach has distinct characteristics, strengths, and appropriate applications that must be carefully matched to your research questions.

Quantitative



- Focus:** Measurable variables and numerical data
- Purpose:** Statistical inference, hypothesis testing, prediction
- Methods:** Experiments, surveys, regression analysis, machine learning models
- Strengths:** Generalizability, objectivity, reproducibility, large-scale analysis
- Examples:** Predicting customer churn, testing algorithm performance, measuring correlation between variables

The choice between these approaches should never be arbitrary or based on personal preference alone. Instead, let your research questions guide you: What type of knowledge are you seeking to generate? What kind of data will best answer your questions? What resources and constraints do you face? In data science research, quantitative methods often dominate, but qualitative approaches can provide crucial context for interpreting model outputs, understanding user needs, or exploring the social implications of algorithmic systems.

Qualitative



- Focus:** Meaning, context, and interpretation
- Purpose:** Understanding phenomena, exploring experiences, generating theory
- Methods:** Interviews, focus groups, ethnography, case studies, thematic analysis
- Strengths:** Depth, context, flexibility, uncovering unexpected insights
- Examples:** Understanding user behavior, exploring algorithmic bias impacts, analyzing stakeholder perspectives

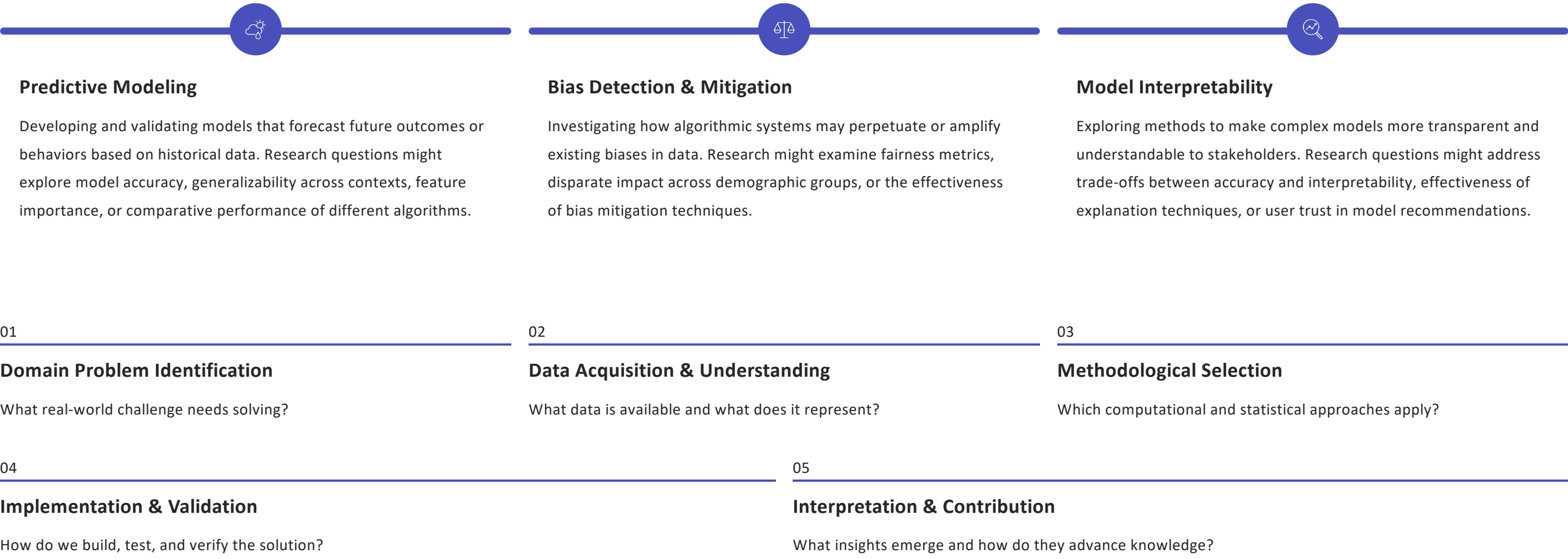
Mixed Methods



- Focus:** Combining interpretation with measurement
- Purpose:** Comprehensive understanding through methodological triangulation
- Methods:** Sequential or concurrent integration of quantitative and qualitative approaches
- Strengths:** Completeness, validation through multiple data sources, addressing complex questions
- Examples:** Model performance metrics + user interviews, survey data + case studies

Research in Data Science: A Unique Context

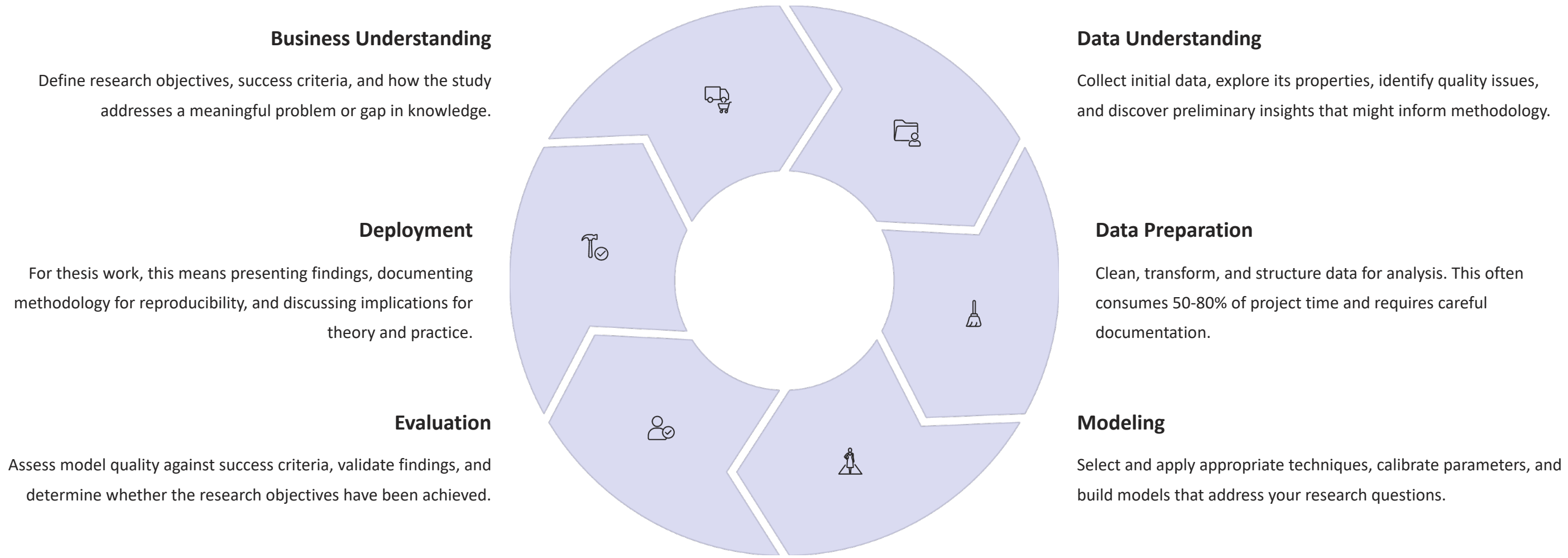
Data science research occupies a distinctive methodological space that blends computational methods, statistical rigor, and domain-specific knowledge. Unlike purely computational research or traditional statistical studies, data science investigations must navigate the intersection of theory, algorithms, and real-world application contexts. This hybrid nature creates both opportunities and challenges for research design.



Effective data science research design must explicitly address the computational pipeline (data preprocessing, feature engineering, model training), the statistical foundation (assumptions, validation strategies, uncertainty quantification), and the application context (stakeholder needs, ethical implications, practical constraints). This multidimensional perspective distinguishes rigorous data science research from mere technical implementation.

Framework Spotlight: CRISP-DM

The Cross-Industry Standard Process for Data Mining (CRISP-DM) has become the most widely adopted framework for structuring data science projects. Developed in the late 1990s by a consortium of industry leaders, CRISP-DM provides a proven, cyclical approach that ensures systematic thinking throughout the research lifecycle. For MSc thesis work, CRISP-DM offers particular value because it enforces documentation, traceability, and iterative refinement—all essential for academic rigor.



The circular nature of CRISP-DM acknowledges that data science research is rarely linear. You may discover data quality issues during modeling that send you back to data preparation, or evaluation results might prompt you to reconsider your business understanding. This iterative flexibility makes CRISP-DM particularly well-suited to the exploratory nature of thesis research, where unexpected findings often lead to the most valuable insights. When documenting your methodology chapter, explicitly mapping your workflow to CRISP-DM stages demonstrates methodological sophistication and helps reviewers understand your systematic approach.

Framework Spotlight: KDD & SEMMA

While CRISP-DM dominates in practice, two other frameworks offer valuable perspectives for structuring data science research: Knowledge Discovery in Databases (KDD) and SEMMA. Understanding these alternatives helps you appreciate different philosophical approaches to the research process and may better suit certain types of investigations.

KDD (Knowledge Discovery in Databases)



Developed in academic circles in the mid-1990s, KDD emphasizes the **discovery-driven** nature of data science research. It conceptualizes the process as mining databases for previously unknown patterns and relationships.

Key Stages:

- 1. **Selection:** Identifying relevant data sources and creating target datasets
- 2. **Preprocessing:** Cleaning and handling missing values
- 3. **Transformation:** Feature engineering and dimensionality reduction
- 4. **Data Mining:** Applying algorithms to discover patterns
- 5. **Interpretation/Evaluation:** Assessing and presenting discovered knowledge

Best For: Exploratory research where you're searching for unexpected patterns or generating hypotheses from large datasets without predetermined models.

SEMMA



Created by SAS Institute, SEMMA provides a more **statistically-oriented** framework that emphasizes rigorous sampling and model assessment. Its acronym captures five core phases.

Key Stages:

- 1. **Sample:** Creating representative training and validation datasets
- 2. **Explore:** Visualizing and understanding variable relationships
- 3. **Modify:** Selecting, transforming, and creating variables
- 4. **Model:** Building and comparing multiple models
- 5. **Assess:** Evaluating model performance and reliability

Best For: Confirmatory research with clear hypotheses, particularly when statistical rigor in sampling and validation is paramount.

Choosing a Methodology: The Decision Matrix

Selecting the right methodology requires systematic thinking about the alignment between your research goals, the nature of your data, and the analytical techniques at your disposal. The decision isn't about choosing the "best" method in absolute terms, but rather identifying the most appropriate method for your specific research context. This alignment is what transforms a collection of analytical techniques into a coherent research design.

	Predictive Goals Objective: Forecast future outcomes or classify instances with high accuracy Data Type: Labeled historical data with clear outcome variables Appropriate Methods: Regression analysis (linear, logistic, polynomial), neural networks, random forests, gradient boosting, support vector machines Example Question: "Can we predict which patients are at highest risk for hospital readmission within 30 days?"
	Exploratory Goals Objective: Discover hidden patterns, structures, or relationships in data without predefined outcomes Data Type: Unlabeled data or data where relationships are unknown Appropriate Methods: Clustering (k-means, hierarchical, DBSCAN), dimensionality reduction (PCA, t-SNE), association rule mining, natural language processing for topic modeling Example Question: "What distinct customer segments emerge from our transaction data, and what characterizes each segment?"
	Explanatory Goals Objective: Understand causal relationships or mechanisms underlying observed phenomena Data Type: Data with potential confounding variables requiring careful control Appropriate Methods: Regression with controls, propensity score matching, difference-in-differences, structural equation modeling, mediation analysis Example Question: "How does personalized recommendation system usage affect customer purchasing behavior, controlling for baseline preferences?"
	Evaluative Goals Objective: Assess performance, effectiveness, or quality of systems, algorithms, or interventions Data Type: Comparative data from multiple models, systems, or experimental conditions Appropriate Methods: A/B testing, cross-validation, benchmark comparisons, sensitivity analysis, fairness metric evaluation Example Question: "Which machine learning algorithm performs best for detecting fraudulent transactions across different fairness metrics?"

When making methodological decisions, consider these critical factors: (1) **Research question alignment**—does this method directly address what you’ are asking? (2) **Data characteristics**—does your data meet the assumptions and requirements of this method? (3) **Interpretability needs**—do stakeholders need to understand how results were produced? (4) **Resource constraints**—do you have sufficient computational resources, time, and expertise? (5) **Validation requirements**—can you adequately test and validate results from this method? A clear decision matrix that maps these factors to potential methods will strengthen your methodology justification and help you defend your choices during thesis review.

Conceptual vs. Theoretical Framework

Many students confuse conceptual and theoretical frameworks, but understanding the distinction is crucial for articulating your research design clearly. Both frameworks serve to organize your thinking and guide your investigation, but they operate at different levels of abstraction and serve different purposes in your thesis.

Theoretical Framework

A theoretical framework is **grounded in established theories** from your discipline or related fields. It provides the academic foundation and broader intellectual context for your work by connecting your research to existing bodies of knowledge. When you invoke a theoretical framework, you're positioning your study within ongoing scholarly conversations.

Characteristics:

- Based on peer-reviewed theories and models
- Provides conceptual lenses for interpretation
- Generalizable across multiple studies
- References established literature extensively

Examples in Data Science Research:

- Information Processing Theory (for recommender systems)
- Technology Acceptance Model (for AI adoption studies)
- Social Capital Theory (for network analysis)
- Prospect Theory (for decision-making models)

Conceptual Framework

A conceptual framework is a **visual or narrative structure specific to your study**. It shows how you conceptualize the relationships between variables, constructs, or components in your particular research context. Think of it as your custom map of the territory you're exploring.

Characteristics:

- Tailored to your specific research questions
- Often represented through diagrams with boxes and arrows
- Shows assumed relationships between elements
- May draw on multiple theories or be atheoretical

Example Components:

- Independent variables (predictors, features)
- Dependent variables (outcomes, targets)
- Mediating variables (mechanisms)
- Moderating variables (conditions that affect relationships)
- Control variables (factors to account for)

Data Quality Dimensions

Completeness, accuracy, consistency



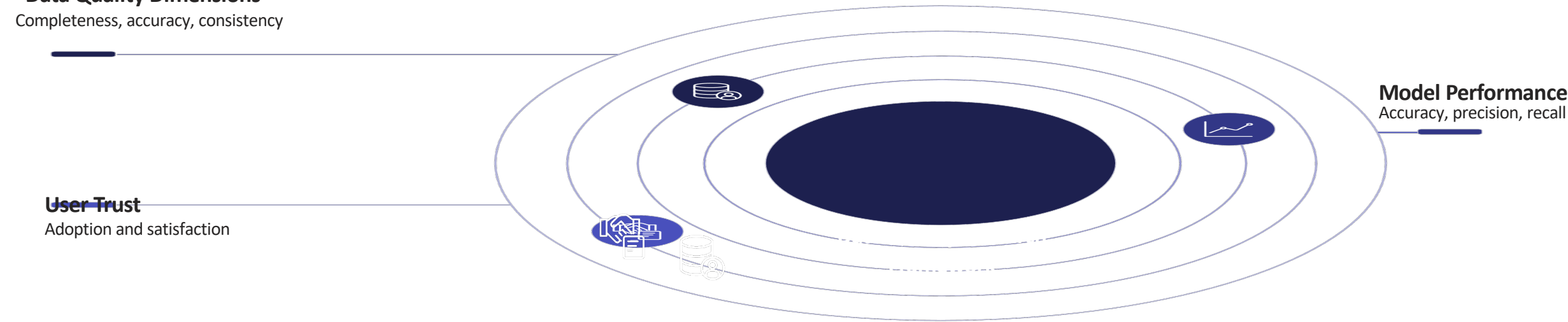
User Trust

Adoption and satisfaction



Model Performance

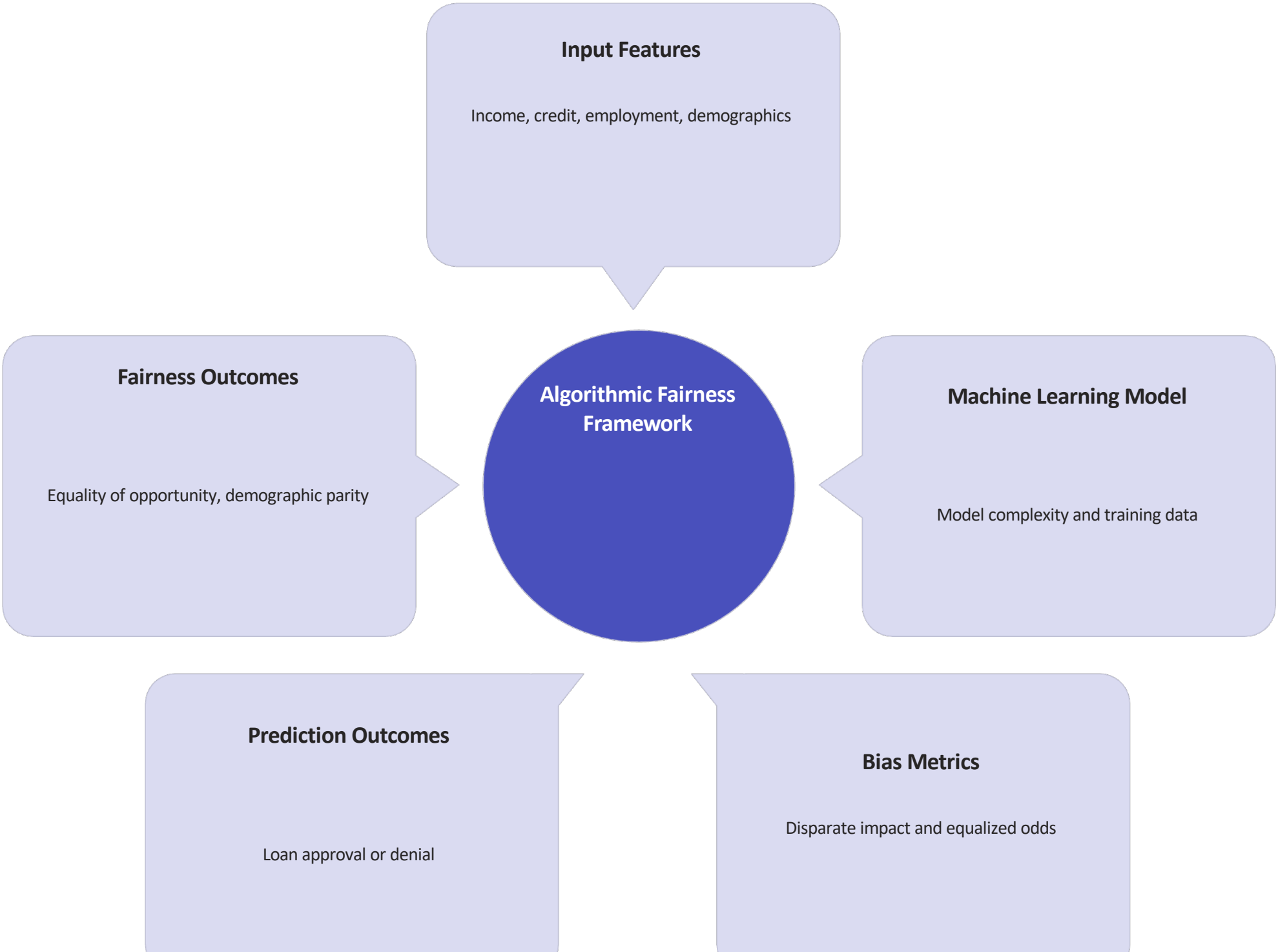
Accuracy, precision, recall



In practice, your conceptual framework often operationalizes your theoretical framework—it shows how abstract theoretical concepts will be measured, modeled, or analyzed in your specific study. For example, if you're using Information Processing Theory (theoretical) to study recommender systems, your conceptual framework might show how information overload (theoretical concept) translates into specific metrics like number of recommendations, user click-through rates, and decision time (operational measures). Strong thesis work typically includes both: a theoretical framework that grounds your work intellectually and a conceptual framework that makes your specific research design transparent and testable.

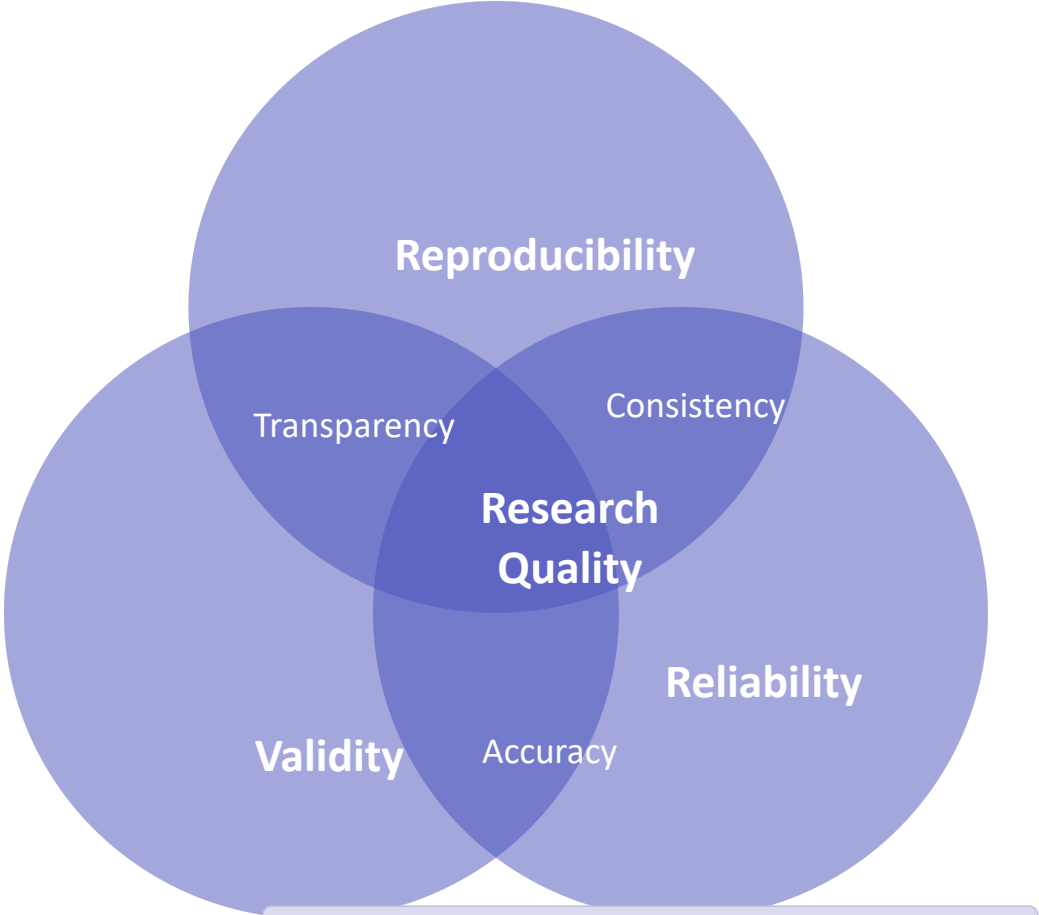
Example: Conceptual Framework in a Data Science Thesis

Let's examine a concrete example to illustrate how conceptual frameworks guide data science research design. Consider a thesis investigating how algorithmic bias in loan approval systems affects outcomes across demographic groups.



Validity, Reliability, and Reproducibility

These three pillars form the foundation of research quality and are non-negotiable in rigorous academic work. While they share the goal of ensuring trustworthy findings, each addresses a distinct dimension of research quality that must be explicitly considered in your design.



Validity: Measuring What Matters

Definition: The extent to which your research design actually investigates what it claims to investigate. Does your study truly measure or test the concepts in your research questions?

Types of Validity:

- **Internal Validity:** Can you confidently attribute observed effects to your variables of interest rather than confounding factors?
- **External Validity:** Do your findings generalize beyond your specific sample or context?
- **Construct Validity:** Do your operational measures accurately represent the theoretical constructs you're studying?
- **Conclusion Validity:** Is your statistical or analytical reasoning sound?

Design Strategies: Use validated measurement instruments, control for confounds, employ appropriate sampling strategies, conduct sensitivity analyses, and clearly define constructs operationally.

Reliability: Consistency of Results

Definition: The degree to which your research produces stable and consistent results when repeated under similar conditions. Would you get essentially the same findings if you ran the study again?

Types of Reliability:

- **Test-Retest Reliability:** Consistency across time points
- **Inter-Rater Reliability:** Agreement between different coders or evaluators (important for qualitative work)
- **Internal Consistency:** Coherence among multiple measures of the same construct

Design Strategies: Set random seeds for reproducible randomization, use cross-validation to assess model stability, document all preprocessing decisions, calculate reliability coefficients (e.g., Cronbach's alpha, Cohen's kappa), and test models on multiple data splits.

Reproducibility: Enabling Verification

Definition: The ability of other researchers to replicate your study and arrive at consistent conclusions using your documented methods and, ideally, your original data and code.

Levels of Reproducibility:

- **Computational Reproducibility:** Others can re-run your code and get identical numerical results
- **Empirical Reproducibility:** Others can collect new data following your methods and test your hypotheses
- **Statistical Reproducibility:** Others analyzing your data reach the same statistical conclusions

Design Strategies: Use version control (Git), document dependencies and environments (requirements.txt, Docker), share code and data (GitHub, OSF), provide detailed method descriptions, and follow reporting standards (TRIPOD for prediction models, PRISMA for reviews).

These three quality dimensions interact and reinforce each other. A valid measure that's unreliable (random noise) isn't useful. A reliable measure that's invalid (consistently measuring the wrong thing) misleads. Reproducible results from an invalid design simply reproduce errors. Your research design must explicitly address all three through methodological choices, documentation practices, and transparency about limitations. Modern data science makes reproducibility particularly critical and attainable—leverage tools like Jupyter notebooks, Docker containers, and data repositories to make your work fully reproducible from raw data to final figures.

Ethical Dimensions in Research Design

Ethics isn't an afterthought or a bureaucratic hurdle—it's a fundamental dimension of research design that shapes every methodological decision. In data science research particularly, where you're often working with human data and building systems that affect people's lives, ethical considerations must be woven into your design from the very beginning. Failure to address ethics adequately can invalidate your entire study, regardless of technical sophistication.

Informed Consent & Data Privacy

If your research involves human participants or their data, you must obtain meaningful informed consent. This means participants understand what data you're collecting, how it will be used, what risks are involved, and that they can withdraw at any time. For secondary data analysis (using existing datasets), verify that original data collection included appropriate consent for your intended use.

Key Considerations: Data anonymization techniques (k-anonymity, differential privacy), secure storage and transmission, compliance with regulations (GDPR, HIPAA, FERPA), and transparency about data retention and sharing plans.

Algorithmic Fairness & Bias Mitigation

Machine learning models can perpetuate or amplify societal biases present in training data. Your research design should explicitly address how you'll measure and mitigate bias, particularly when working with data involving protected characteristics (race, gender, age, disability status).

Key Considerations: Choice of fairness metrics (demographic parity, equalized odds, predictive parity), representation of minority groups in datasets, bias auditing procedures, and documentation of trade-offs between different fairness criteria and accuracy.

Research Integrity & Avoiding Data Misuse

Ethical research means resisting the temptation to "torture the data until it confesses"—manipulating analyses until you get statistically significant or publishable results. This includes avoiding selective reporting, p-hacking, and overfitting models to validation data.

Key Considerations: Pre-registering analysis plans when possible, transparently reporting null results, acknowledging limitations honestly, and avoiding cherry-picking results that support desired conclusions.

Institutional Review Board (IRB) Process

Most universities require IRB approval for research involving human subjects. Even if you're analyzing secondary data or publicly available datasets, consult with your IRB early in your design process. They can help you identify ethical considerations you might have missed and ensure you meet regulatory requirements.

Key Considerations: Determining whether your study requires full board review, expedited review, or is exempt; preparing thorough protocol descriptions; obtaining required certifications (e.g., CITI training); and budgeting time for the review process (often 4-8 weeks).

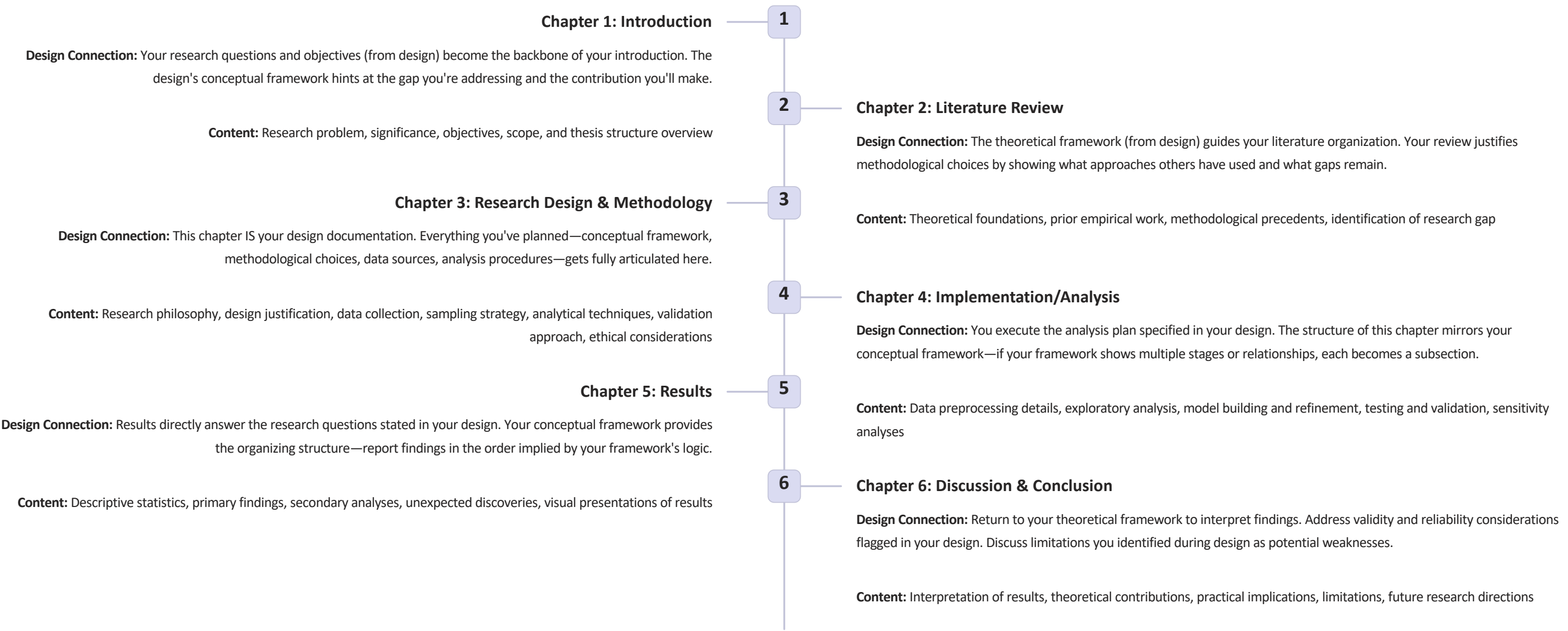


Practical Tip: Start the IRB process as soon as you have a clear research design—delays in ethical approval are one of the most common causes of thesis timeline extensions. Many students underestimate how long ethics review takes or assume their secondary data analysis is automatically exempt. When in doubt, consult your IRB office early.

Ethical research design also means considering the broader societal implications of your work. If you're building a predictive model, who might be harmed if it makes errors? If you're analyzing social media data, might your findings be misused to manipulate or surveil people? If you're recommending a new algorithm, what are the environmental costs of its computational requirements? These questions don't have simple answers, but acknowledging them demonstrates intellectual maturity and positions you as a responsible researcher who thinks beyond technical metrics to real-world impacts.

Mapping Research Design to MSc Thesis Structure

Your research design doesn't exist in isolation—it's the organizational logic that connects every chapter of your thesis into a coherent narrative. Understanding how design decisions flow through your document helps you maintain consistency and makes writing much more manageable. Early clarity in your design saves countless hours of confusion later.



Notice the throughline: research questions from your design anchor Chapter 1, guide your literature review in Chapter 2, get operationalized in Chapter 3, drive analysis in Chapter 4, structure results in Chapter 5, and are answered in Chapter 6. Similarly, your conceptual framework appears first in Chapter 3, gets tested in Chapter 4, generates findings in Chapter 5, and informs interpretation in Chapter 6. This alignment isn't coincidental—it's what makes a thesis feel coherent rather than like a collection of disconnected chapters. When you invest time in a solid research design up front, you're essentially creating an outline for your entire thesis. Each chapter then becomes an elaboration of one component of your design rather than a separate challenge to tackle from scratch.

Case Study: Predictive Analytics in Healthcare

Let's examine a complete research design for a realistic data science thesis: predicting hospital readmissions within 30 days of discharge. This case study illustrates how design decisions interconnect and how abstract principles become concrete methodological choices.

Research Problem

Hospital readmissions are costly and often preventable. Can machine learning models identify high-risk patients to enable targeted interventions?

Research Questions

1.
- Which patient and clinical factors most strongly predict 30-day readmission?
2.
- How do different ML algorithms compare in predictive performance?
3.
- Can model predictions support clinically actionable risk stratification?

Design Classification

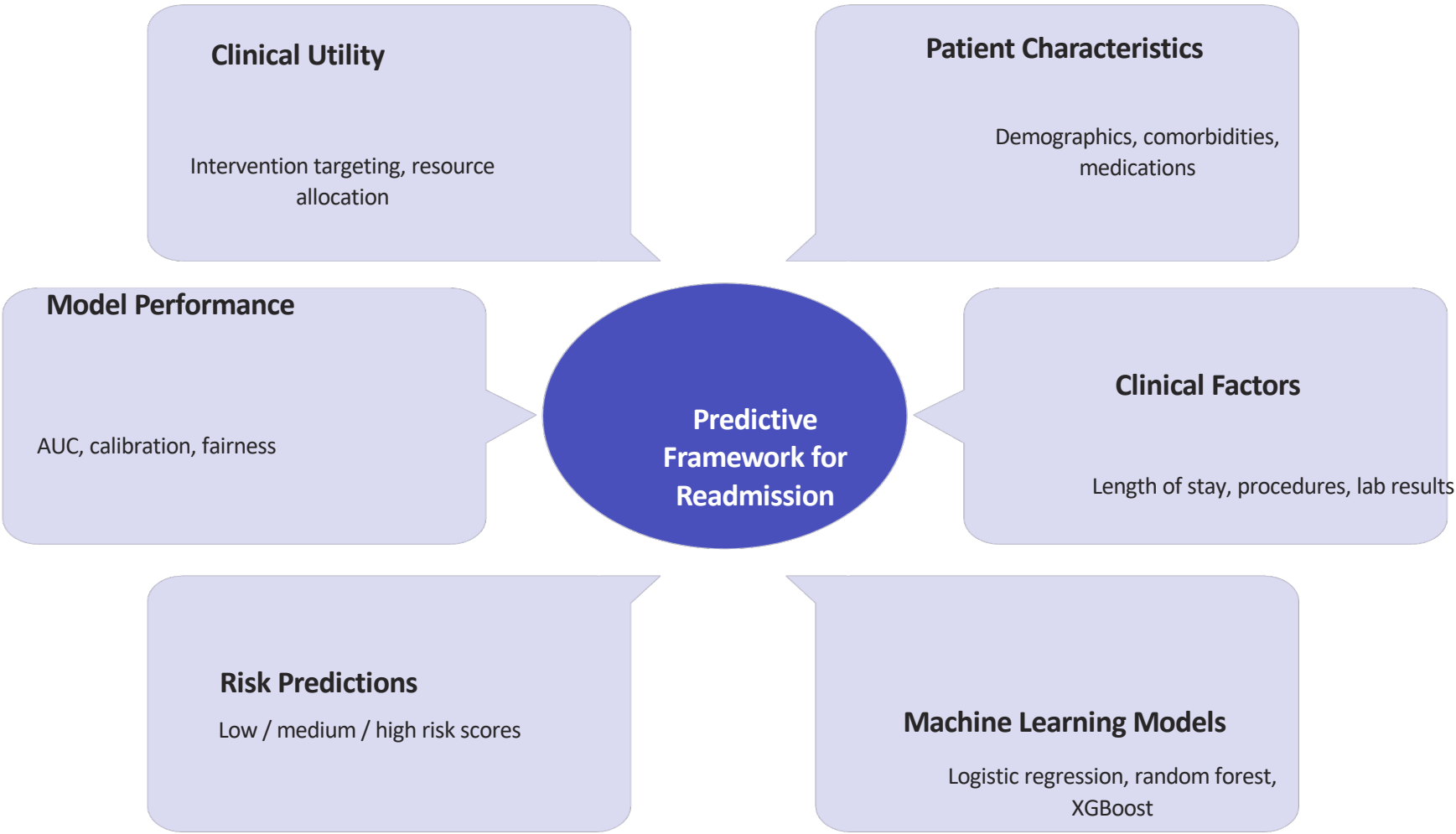
Approach: Quantitative, predictive modeling study

Framework: CRISP-DM provides the overall structure

Research Type: Applied research with evaluation component

Theoretical Grounding

- Clinical Decision Support Theory:** How information systems enhance medical decision-making
- Risk Stratification Literature:** Prior work on patient risk assessment and care m



01	02	03
Data Source	Sampling Strategy	Feature Engineering
Electronic health records (EHR) from a regional hospital system, 2018-2022. N = 50,000 admissions across multiple clinical departments. Includes demographics, diagnoses (ICD-10), procedures, medications, and outcomes.	Include all adult medical/surgical admissions. Exclude: pediatric patients, psychiatric admissions, patients who died during hospitalization, and incomplete records. Training/validation/test split: 60%/20%/20% with temporal validation (test set from most recent year).	Create features from structured EHR data: comorbidity indices (Charlson, Elixhauser), medication counts by class, procedure complexity scores, laboratory value abnormalities, prior utilization history. Handle missing data through multiple imputation.
04	05	
Modeling Approach	Evaluation Metrics	
Compare three algorithms: logistic regression (baseline/interpretability), random forest (handling non-linearity), XGBoost (performance). Tune hyperparameters via grid search with cross-validation on training set. Evaluate on held-out test set only once.	Performance: AUC-ROC, precision-recall curves, calibration plots. Fairness: Stratified performance by age, gender, race/ethnicity. Clinical Utility: Net benefit analysis, decision curve analysis to assess practical value.	
Ethical Considerations	Validity Threats	Expected Contributions
- IRB approval obtained before data access - Retrospective analysis of de-identified data (HIPAA compliant) - Fairness analysis to detect disparate impact - No deployment during research phase—findings inform future system development	- Internal: Confounding by indication, measurement error in EHR data - External: Single health system may limit generalizability to other populations - Mitigation: Sensitivity analyses, temporal validation, comparison to published benchmarks	- Methodological: Comparative evaluation of algorithms in healthcare context - Practical: Identification of actionable risk factors for intervention targeting - Theoretical: Evidence on clinical decision support model requirements
This design demonstrates several hallmarks of strong thesis work: clear research questions that build on each other, explicit methodological justification (why these algorithms? why these metrics?), attention to both performance and fairness, and acknowledgment of limitations up front. The CRISP-DM framework provides structure while the conceptual framework makes assumptions and relationships explicit. Most importantly, every design element connects logically to the research questions—nothing is included just because it's sophisticated or because "everyone does it this way." This intentionality is what separates rigorous research from mere technical demonstration.		

Common Design Pitfalls

Even experienced researchers fall into predictable traps during research design. Being aware of these common pitfalls helps you avoid costly mistakes that might force you to redesign your study midway through your thesis. Here are the most frequent design errors and how to avoid them.

✖ What NOT to Do

Choosing Methods Before Questions

"I want to use deep learning" shouldn't come before "What am I trying to find out?" Method fascination leads to solutions searching for problems. Always start with research questions, then select methods that best address them.

Ignoring Data Availability

Designing a study around data you assume exists but haven't verified access to. By the time you discover the data isn't available or doesn't contain necessary variables, you've wasted weeks. Verify data access *before* finalizing your design.

Skipping Reproducibility Planning

Assuming you'll "document everything later" leads to irreproducible research. Set up version control, documentation practices, and computational environment management from day one. Future you (and your reviewers) will be grateful.

Overlooking Ethical Approval

Starting data collection before IRB approval or assuming secondary data analysis is automatically exempt. Ethics review can take months—ignoring it derails timelines and may invalidate your work entirely.

Underestimating Data Preparation

Allocating 90% of timeline to modeling and 10% to data cleaning, when reality is inverted. Data wrangling typically consumes 60-80% of project time. Plan accordingly and build in buffer time.

Complexity for Complexity's Sake

Using the most sophisticated methods available rather than those best suited to your question. Simple methods that work and can be interpreted often beat complex black boxes that perform marginally better.

✔ Best Practices

Questions Drive Methods

Articulate research questions first, then identify methods that best address them. Justify each methodological choice explicitly: "I'm using X because it allows me to Y, which is necessary to answer my research question about Z."

Verify Data Early

Obtain sample data or detailed documentation before committing to a design. Verify: variable availability, data quality, sample size, temporal coverage, and access permissions. If data doesn't support your question, pivot early.

Build Reproducibility Infrastructure

From day one: use Git for version control, maintain requirements.txt or environment.yml for dependencies, write README files, comment code thoroughly, and organize project directories logically. Make reproducibility automatic, not afterthought.

Start Ethics Process Immediately

Consult with your IRB office as soon as you have a draft research design. Submit applications early in your timeline, even before data access is finalized. Plan for 4-8 week review cycles and potential revisions.

Allocate Time Realistically

Use 60/40 rule: 60% of time for data acquisition, cleaning, and exploration; 40% for modeling and analysis. Build in 20% buffer time for unexpected issues. Data preparation always takes longer than you think.

Appropriate Sophistication

Choose methods that match your question's complexity, data characteristics, and interpretability needs. Start simple (baseline models), then increase complexity only if necessary and justified. Occam's Razor applies to research design.

🚩 **Red Flag Check:** If you can't explain your methodological choices to a non-expert without resorting to "because it's cutting-edge" or "everyone uses it," you may not understand your own design well enough. The ability to justify choices simply is a sign of deep understanding.

Interactive Workshop: Method Matchmaker

Let's test your understanding with a practical exercise. For each research scenario below, identify the most appropriate design approach and methodological choices. Consider the research question, data type, desired output, and constraints.

1

Scenario: Mental Health Risk Prediction

Context: A university counseling center wants to identify students at elevated risk for mental health crises based on academic performance, attendance, and self-reported wellness surveys.

Available Data: Structured academic records (grades, course load, attendance), Likert-scale wellness surveys (stress, sleep, social connection), demographic information. N = 8,000 students, 3 years of data.

Constraints: Results must be interpretable to non-technical counselors. Predictions will inform outreach, not automated decisions. Strong ethical and privacy requirements.

Your Task: What research design, methodology, and evaluation approach would you recommend?

2

Scenario: E-Commerce Product Recommendation

Context: An online retailer wants to understand whether collaborative filtering or content-based recommendation systems better serve different customer segments.

Available Data: Purchase history (item, timestamp, price), product metadata (category, brand, attributes), customer demographics, clickstream data. N = 100,000 customers, 1 million transactions.

Constraints: Must compare multiple approaches systematically. Need to balance recommendation accuracy with novelty/diversity. Computational resources are limited.

Your Task: What research design, methodology, and evaluation approach would you recommend?

3

Scenario: Algorithm Bias in Hiring

Context: A tech company uses ML for resume screening and wants to audit whether their system exhibits bias against protected groups.

Available Data: Historical hiring data (resumes, screening decisions, interview outcomes, hires), candidate demographics (inferred from names, optional self-identification), job descriptions. N = 50,000 applications over 2 years.

Constraints: Must assess fairness across multiple protected characteristics. Need both quantitative bias metrics and qualitative understanding of mechanisms. Results will inform policy decisions.

Your Task: What research design, methodology, and evaluation approach would you recommend?

4

Scenario: Customer Segmentation

Context: A retail bank wants to discover natural customer segments for personalized marketing but doesn't have predefined categories.

Available Data: Transaction history (amount, frequency, merchant categories), account balances, product holdings (checking, savings, loans, credit cards), demographics. N = 250,000 customers, 5 years of data.

Constraints: Exploratory analysis—no predetermined segments. Results must be actionable for marketing team. Need to identify both segment characteristics and appropriate number of segments.

Your Task: What research design, methodology, and evaluation approach would you recommend?

5

Scenario: Explainable AI for Medical Diagnosis

Context: Researchers are developing a system to detect diabetic retinopathy from retinal images but physicians need to understand how the system makes decisions.

Available Data: 10,000 labeled retinal images (graded by ophthalmologists on 5-point severity scale), patient metadata (age, diabetes duration, HbA1c levels), high-resolution images require substantial preprocessing.

Constraints: Must achieve clinical-grade accuracy while maintaining interpretability. Physicians skeptical of black-box AI. Results should identify both what the model sees and how confident it is.

Your Task: What research design, methodology, and evaluation approach would you recommend?

Suggested Approach for Each Scenario

Scenario 1 (Mental Health): Quantitative predictive study using CRISP-DM framework. Start with interpretable models (logistic regression, decision trees) to identify risk factors. Use fairness metrics to ensure no disparate impact. Cross-validation with temporal splits. Emphasize recall over precision (better to over-identify risk than miss cases). IRB approval essential.

Scenario 2 (Recommendations): Mixed methods approach combining quantitative algorithm comparison with qualitative user feedback. Implement multiple algorithms, compare via A/B testing or offline evaluation (precision@k, NDCG, coverage, diversity metrics). Segment analysis b y customer characteristics. Survey subset of users on satisfaction and novelty perceptions.

Design Tools for Students

Modern research design benefits from a rich ecosystem of digital tools that can help you organize your thinking, visualize frameworks, manage references, and collaborate with advisors. Investing time in learning these tools early pays dividends throughout your thesis journey.



Mendeley/Zotero

Purpose: Reference management and framework documentation

Key Features: Browser plugin for one-click citation capture, PDF annotation with cloud sync, automatic bibliography generation in any citation style, tagging for organizing sources by theme

Use Case: Store and organize literature that informs your theoretical framework, tag sources by methodology to identify patterns, export citations directly into your thesis



Lucidchart / Draw.io

Purpose: Creating conceptual frameworks and process diagrams

Key Features: Drag-and-drop interface for flowcharts and diagrams, extensive shape libraries, real-time collaboration, export to multiple formats (PNG, PDF, SVG)

Use Case: Design your conceptual framework with boxes and arrows, map CRISP-DM stages to your specific workflow, create data pipeline visualizations

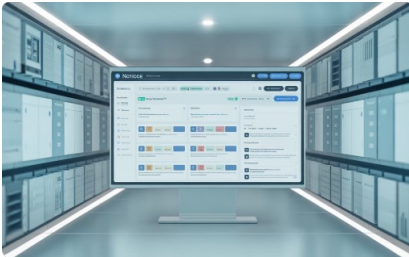


Miro

Purpose: Visual brainstorming and framework iteration

Key Features: Infinite canvas for free-form diagramming, sticky notes and mind maps, templates for various frameworks, excellent for remote collaboration with advisors

Use Case: Brainstorm research questions and map relationships, create literature synthesis boards, iterate on framework designs before formalizing

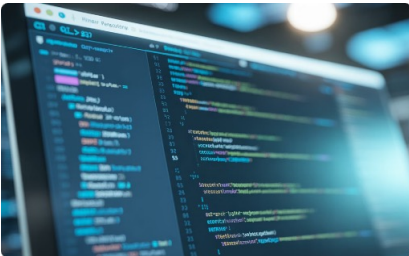


Notion

Purpose: Integrated workspace for design planning and documentation

Key Features: Flexible database and note-taking system, kanban boards for task management, embedding capabilities for diagrams and references, templates for research planning

Use Case: Maintain a research design document that evolves with your thinking, track methodological decisions and justifications, organize weekly meeting notes with advisors



GitHub / GitLab

Purpose: Version control and reproducibility infrastructure

Key Features: Git-based version control, issue tracking for managing analysis tasks, README documentation, automated testing and continuous integration capabilities

Use Case: Version control for analysis code, document analysis decisions in commit messages, share code with advisors for review, prepare for open science publishing

Additional Specialized Tools



ChatGPT / Claude

Purpose: Drafting and brainstorming assistant

Key Features: Natural language interaction for exploring ideas, assistance with technical writing, coding help, literature synthesis support

Use Case: Draft methodology sections and get feedback, brainstorm methodological alternatives, clarify statistical concepts, generate code templates (always verify!)—but never use for literature content without verification

Tool Selection Tips

Don't feel obligated to use every tool listed here. Instead:

- Start with 2-3 core tools (reference manager, diagramming tool, note-taking system) and add others as needs arise
- Choose tools your advisor is familiar with to facilitate collaboration
- Prioritize tools that enhance reproducibility and documentation
- Learn tools incrementally—basic competence in core features beats advanced features you'll never use
- Avoid tool-hopping mid-thesis; commit to a toolchain early and stick with it

Pro Tip: Set up your tool ecosystem in the first week of thesis work, before diving into literature or data. Having infrastructure ready makes subsequent work more efficient and prevents the "I'll organize this later" trap that leads to chaos.

- **Obsidian:** Knowledge graph tool for connecting ideas across literature, frameworks, and findings
- **Mendeley:** Alternative reference manager with PDF reader and citation plugin
- **Overleaf:** Collaborative LaTeX editor for thesis writing (particularly useful for equations and technical content)
- **Jupyter Notebooks:** Literate programming environment that combines code, visualizations, and narrative—ideal for computational research
- **Docker:** Containerization for ensuring your computational environment is reproducible

Assignment & Deliverable

To solidify your understanding of research design principles, you will complete a practical assignment that forms the foundation for your thesis methodology chapter. This exercise helps you articulate your design decisions clearly and receive early feedback before committing significant time to implementation.

Research Design Proposal

Length: 1-2 pages (approximately 500-750 words)

Format: PDF or Word document, 12pt font, 1.5 line spacing

Due Date: tba

Submission Method: send to me

Required Components

01	02	03
Research Questions & Objectives State 2-3 specific research questions that your thesis will address. These should be clear, focused, and answerable through empirical investigation. Briefly explain why these questions matter and what gap they address.	Methodological Approach Identify your overall approach (quantitative, qualitative, or mixed methods) and justify why this approach best suits your research questions. Specify which data science framework you'll apply (CRISP-DM, KDD, SEMMA, or hybrid) and why.	Data & Analysis Plan Describe your data source(s), including type of data, approximate sample size, and key variables. Outline your planned analytical techniques (e.g., specific ML algorithms, statistical tests, qualitative coding methods) and explain why these techniques are appropriate.
04	05	
Validity & Ethics Address how you'll ensure validity and reproducibility (e.g., validation strategies, documentation practices). Note any ethical considerations and whether IRB approval will be required. Evaluation Criteria Your proposal will be assessed on: Clarity (30%): Are your research questions and approach clearly articulated? Alignment (30%): Do your methodological choices logically connect to your research questions? Feasibility (20%): Is your design realistic given typical MSc timeline and resource constraints? Rigor (20%): Have you considered validity, ethics, and reproducibility? This is a formative assessment—the goal is feedback to strengthen your thesis design, not to penalize imperfect early drafts. Be honest about uncertainties and open to revising your plan based on feedback.	Conceptual Framework (Optional) If relevant to your study, include a simple diagram showing the relationships between key variables or constructs. This can be hand-drawn and photographed or created using diagramming software. Timeline & Next Steps Week 5, Day 1-2 Brainstorm research questions and review available data sources Week 5, Day 3-4 Draft methodology section and create conceptual framework Week 5, Day 5-6 Revise for clarity, check alignment, prepare submission Week 5, Day 7 Submit proposal before deadline Week 6 Receive feedback, schedule meeting with advisor if needed Week 7+ Incorporate feedback into full thesis methodology chapter	
Advisor Consultation: Before finalizing your proposal, schedule a brief meeting with your thesis advisor to discuss your ideas. Their early input can save you from pursuing infeasible directions. Bring a rough draft or at minimum, a bulleted outline of your thinking.		
IJA		

Recap & Reflection

We have covered substantial ground this week, moving from abstract principles to concrete design decisions. Let's consolidate the key takeaways and prepare for what's ahead in your research journey.

Design as Backbone

Research design is the structural foundation that holds your entire thesis together. It's not bureaucratic box-checking—it's the strategic thinking that ensures every methodological choice serves your research questions and every component connects logically to the others.

Alignment is Everything

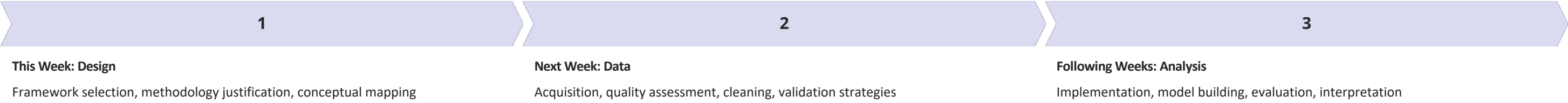
The through-line connecting Why → What → How is what distinguishes coherent research from scattered exploration. Your theoretical framework informs your conceptual framework, which guides your methodology, which structures your analysis, which addresses your research questions. This alignment is non-negotiable.

Frameworks Guide, Don't Constrain

CRISP-DM, KDD, SEMMA, and similar frameworks provide structure and ensure systematic thinking, but they're tools to serve your research, not rigid templates to force your work into. Adapt them to your needs while maintaining their core logic.

From Design to Execution

With a solid research design in hand, you're ready to transition from planning to doing. Next week, we'll dive into the practical realities of working with data—how to acquire, clean, validate, and prepare data for analysis. You'll learn that data preparation isn't just technical work; it's an extension of your research design where decisions about handling missing values, outliers, and transformations directly impact what you can learn from your analysis.



Reflection Prompts

Take a few minutes to consider these questions as you solidify your understanding:

- Can I explain my methodological choices to someone outside my field in non-technical language? If not, do I truly understand them?
- What assumptions am I making in my research design, and have I made them explicit?
- Which design decisions am I most confident about? Which ones would benefit from advisor input?
- How will I know if my design is working as I implement it? What would signal that I need to pivot?
- What's the single biggest threat to the validity of my findings, and how can I mitigate it?

"Research design is where the rubber meets the road between interesting questions and answerable questions. Your job is to build the road that gets you from one to the other."

— Anonymous Research Methods Professor

Remember: investing time in robust research design isn't about perfection—it's about clarity, intentionality, and setting yourself up for thesis success. The decisions you make now will shape every subsequent phase of your work. Take them seriously, seek feedback early, and don't be afraid to revise your design as you learn more. Research is inherently iterative, and thoughtful adaptation is a sign of intellectual maturity, not weakness.

Next week: From Design to Data – Acquisition, Validation, and Quality Assurance. See you then!